QUESTION ANALYSIS

EEE443

Power Plant Engineering

Question Analysis

Sessions 2020–21 *1 papers, 24 questions, 31 formulas*

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1. At a glance:

This analysis covers **1 paper** (2020-21), with 24 questions and 150 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

With one paper, no trend can be read. Use this as a map of the topics.

The five topics with the most marks:

Topic	Marks	Share	Questions	Top subtopics (marks)
Steam power plant	30	20 %	5	Boiler efficiency (8); Layout (7); Improvements (5)
Load curves and factors	29	19 %	5	Factors (15); Base and peak load (10); Variable load (4)
Hydro plants	20	13 %	3	Turbines (8); Civil works (6); Merits (6)
Economic load sharing	18	12 %	2	Incremental rates (10); Two units (8)
Tariffs	14	9 %	2	Two-part (9); Flat and meter rates (5)

In the topic pages, a formula name in bold is a key formula, the number after a subtopic is its marks offered, and a blue symbol in an equation links to its meaning in the symbol list, where the arrow leads back to the equation.

2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

Topic and subtopic	2020-21	All
Load curves and factors	**29**	**29**
Factors	**15**	15
Base and peak load	**10**	10
Variable load	**4**	4
Economics of generation	**12**	**12**
Depreciation	**7**	7
Cost per kWh	**5**	5
Tariffs	**14**	**14**
Two-part	**9**	9
Flat and meter rates	**5**	5
Economic load sharing	**18**	**18**
Incremental rates	**10**	10
Two units	**8**	8
Fuels and handling	**11**	**11**
Pulverisation	**6**	6
Characteristics	**5**	5
Steam power plant	**30**	**30**
Boiler efficiency	**8**	8
Layout	**7**	7
Improvements	**5**	5
Boilers	**5**	5
Superheaters and economisers	**5**	5
Steam turbines	**6**	**6**
Impulse and reaction	**6**	6
Combined cycle plants	**5**	**5**
Operation	**5**	5

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Topic and subtopic	2020-21	All
Hydro plants	**20**	**20**
Turbines	**8**	8
Civil works	**6**	6
Merits	**6**	6
Nuclear plants	**5**	**5**
Reactors	**5**	5

3. Priority list:

Must know: Steam power plant (30), Load curves and factors (29), Hydro plants (20), Economic load sharing (18), Tariffs (14), Economics of generation (12), Fuels and handling (11), Steam turbines (6), Combined cycle plants (5), Nuclear plants (5)

Likely: none

Less often: none

The rule: Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

4. Topic pages:

4.1. Load curves and factors:

29 marks, 5 questions, in 1 of 1 papers. Book: Mehta Ch. 3.

Most often theory (4 of 5). Subtopics: Base and peak load; Factors; Variable load.

Formulas:

Subtopic	Formula	Equation	Used in
Base and peak load (10)	**Average load**	$P_{av} = \dfrac{E}{t}$	20-21 1(d)
	Load factor	$\text{LF} = \dfrac{P_{av}}{P_{max}}$	20-21 1(d)
	Diversity factor	$\text{DF} = \dfrac{\sum P_{max,i}}{P_{max}} \geq 1$	20-21 1(d)
	Demand factor	$\dfrac{P_{max}}{\text{connected load}} \leq 1$	20-21 1(d)
Factors (15)	**Capacity factor**	$\text{CF} = \dfrac{P_{av}}{C} = \dfrac{E}{C\,t}$	20-21 4(c)
	Plant use factor	$\text{UF} = \dfrac{E}{C \times t_{op}}$	20-21 4(a)
	Utility (utilisation) factor	$\dfrac{P_{max}}{C}$	20-21 4(a)
	Reserve capacity	$C - P_{max}$	20-21 4(c)
	Energy at full load as per schedule	$E_{sched} = C\,t_{op} = \dfrac{E}{\text{UF}}$	20-21 4(c)

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Symbols:

P_{av} = average load (kW) ↩

E = energy in the period (kWh) ↩

t = hours in the period (8760 for a year, 24 for a day) ↩

P_{max} = maximum demand on the station (kW) ↩

$P_{max,i}$ = maximum demand of consumer group i ↩

C = plant capacity (rated) ↩

t_{op} = hours the plant was in operation ↩

E_{sched} = energy at full load in the hours the plant ran (kWh) ↩

Questions:

No.	Paper	Marks	Type	Question
Base and peak load				
1	20-21 1(d)	10	theory	What are the requirements of a base load and peak load power plant? The annual peak load on a 30 MW power station is 25 MW. The power station supplies loads having a maximum demands of 10MW, 8.5 MW, 5 MW and 4.5 MW. The annual load factor is 45%. Find: (i) Average load, (ii) Energy supplied every year, (iii) Diversity factor, (iv) Demand/factor. *[sic: should be "Demand factor"]*
Factors				
2	20-21 2(a)	3	theory	What is the importance of load factor and diversity factor?
3	20-21 4(a)	4	theory	Define plant use factor and plant utility factor.
4	20-21 4(c)	8	numerical	A generating station has a maximum demand of 20 MW, a load factor of 60%, a plant capacity factor of 48% and a plant use factor of 80%. Find: (i) The daily energy produced (ii) The reserve capacity of the plant (iii) The maximum energy that could be produced daily if the plant was running all the time (iv) The maximum energy that could be produced daily if the plant was running fully loaded and operating as per schedule.
Variable load				
5	20-21 2(b)	4	theory	Why is the load on a power station variable? What are the effects of variable load on the operation of the power station?

4.2. Economics of generation:

12 marks, 2 questions, in 1 of 1 papers. Book: Mehta Ch. 4.

Most often numerical (1 of 2). Subtopics: Cost per kWh; Depreciation.

Formulas:

Subtopic	Formula	Equation	Used in
Cost per kWh (5)	Energy in a year	$E = C \times \text{LF} \times 8760$	20-21 5(b)
	Annual fixed charges	$(i + d)\,K + W$	20-21 5(b)
	Cost per kWh	$\dfrac{\text{annual fixed charges}}{E} + \text{running cost per kWh}$	20-21 5(b)
Depreciation (7)	Straight line	$D = \dfrac{P - S}{n}$ per year	20-21 6(a)
	Diminishing value	$P(1-r)^n = S \;\Rightarrow\; r = 1 - \left(\dfrac{S}{P}\right)^{1/n}$	20-21 6(a)
	Sinking fund	$q = (P - S)\,\dfrac{i}{(1+i)^n - 1}$	20-21 6(a)

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Symbols:

E = energy generated in a year (kWh) ↩

C = plant capacity (kW) ↩

K = capital cost ↩

i, d = interest and depreciation rates ↩

W = wages, taxes ↩

D = depreciation per year (Tk) ↩

P = initial cost (Tk) ↩

S = salvage value (Tk) ↩

n = useful life (years) ↩

r = fraction of the value left, charged each year ↩

q = annual deposit ↩

i = interest rate ↩

Questions:

No.	Paper	Marks	Type	Question
Cost per kWh				
1	20-21 5(b)	5	numerical	Estimate the generating cost per KWh delivered from a generating station from the following data: Plant capacity = 50 MW. Annual load factor = 40%. Capital cost = 1.2 crores. Annual cost of wages, taxation, etc. = Taka 4 lakhs. Cost of fuel, lubrication, maintenance, etc. = 1.0 paisa/kWh generated. Interest 5% per annum, depreciation 6% per annum of initial value.
Depreciation				
2	20-21 6(a)	7	theory	What are the methods of determining depreciation cost in power plants? Briefly explain one of the methods.

4.3. Tariffs:

14 marks, 2 questions, in 1 of 1 papers. Book: Mehta Ch. 5.

Most often theory (1 of 2). Subtopics: Flat and meter rates; Two-part.

Formulas:

Subtopic	Formula	Equation	Used in
Flat and meter rates (5)	Flat demand rate	$\text{bill} = a \times \text{kW}$	20-21 5(a)
	Straight line meter rate	$\text{bill} = b \times \text{kWh}$	20-21 5(a)
Two-part (9)	**Two-part tariff**	$\text{bill} = A \times \text{kW} + B \times \text{kWh}$	20-21 5(d)
	Demand charge	$A = \dfrac{\text{annual fixed charges}}{\text{aggregate maximum demand of consumers}}$	20-21 5(d)
	Energy charge	$B = \dfrac{\text{annual running charges}}{(1 - \text{loss})\, E_{gen}}$	20-21 5(d)

No.	Paper	Marks	Type	Question

Symbols:

a = Tk per kW connected ↩

b = Tk per kWh ↩

A = demand charge (Tk per kW per year) ↩

B = energy charge (Tk per kWh) ↩

E_{gen} = energy generated (kWh) ↩

Questions:

No.	Paper	Marks	Type	Question
Flat and meter rates				
1	20-21 5(a)	5	theory	Write short notes on the following type of tariffs: (i) Flat demand rate (ii) Straight line meter rate
Two-part				
2	20-21 5(d)	9	numerical	An electric supply company having a maximum load of 50 MW generates 18×10^7 units per annum and the supply consumers have an aggregate demand of 75 MW. The annual expenses including capital charges are: For fuel = Taka 90 lakhs. Fixed charges concerning generation = Taka 28 lakhs. Fixed charges concerning transmission and distribution = Taka 32 lakhs. Assuming 90% of the fuel cost is essential to running charges and the loss in transmission and distribution as 15% of KWh generated, deduce a two part tariff to find the actual cost of supply to the consumers.

4.4. Economic load sharing:

18 marks, 2 questions, in 1 of 1 papers. Book: Nag Ch. 1; Mehta Ch. 3.

Most often derivation (1 of 2). Subtopics: Incremental rates; Two units.

Formulas:

Subtopic	Formula	Equation	Used in
Incremental rates (10)	Heat rate	$\frac{I}{P}$	20-21 2(c)
	Incremental heat rate	$\frac{dI}{dP}$; in money $\frac{dF}{dP}$ (Tk/MWh)	20-21 2(c)
	Economic sharing	$\frac{dI_1}{dP_1} = \frac{dI_2}{dP_2} = \cdots = \lambda,\ \sum P_i = P$	20-21 2(c)
Two units (8)	Two linear units	$P_A = \frac{a_B P + b_B - b_A}{a_A + a_B}$	20-21 2(d)
	Cost of moving load	$\Delta F = \int_{P_1}^{P_2} \frac{dF}{dP}\, dP = \frac{a}{2}(P_2^2 - P_1^2) + b(P_2 - P_1)$	20-21 2(d)

Symbols:

I = heat input (kcal/h) ↩

P = output (MW); total load, fixed (MW) ↩

F = fuel cost (Tk/h)

I_1, I_2 = heat inputs of units 1 and 2 (kcal/h) ↩

P_1, P_2 = outputs of units 1 and 2 (MW); load before and after the move (MW) ↩

λ = common incremental rate ↩

P_i = output of unit i (MW) ↩

P_A = load on unit A (MW) ↩

a_A, b_A = slope and intercept of unit A's incremental cost ↩

a_B, b_B = the same for unit B ↩

ΔF = change of fuel cost (Tk/h) ↩

a, b = slope and intercept of the incremental-cost line ↩

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Questions:

No.	Paper	Marks	Type	Question
Incremental rates				
1	20-21 2(c)	10	derivation	Show that for the economic load sharing, the incremental heat rates are equal.
Two units				
2	20-21 2(d)	8	numerical	The incremental fuel costs for two generating units A and B of a power plant are given by the following equations: $\frac{dF_A}{dP_A} = 0.06P_A + 11.4$, $\frac{dF_B}{dP_B} = 0.07P_B + 10$, where P is in mega-watts and F is in takas per hour. (i) Find the economic loading of the two units when the total load to be supplied by the power station is 150 MW. (ii) Find the loss in fuel cost per hour if the load is equally shared by the two units.

4.5. Fuels and handling:

11 marks, 2 questions, in 1 of 1 papers. Book: Nag Ch. 4, 5.

Most often theory (2 of 2). Subtopics: Characteristics; Pulverisation.

Questions:

No.	Paper	Marks	Type	Question
Characteristics				
1	20-21 1(b)	5	theory	Briefly discuss the characteristics of fuels.
Pulverisation				
2	20-21 6(b)	6	theory	Why is pulverization done in a coal fired power plant? Give description of any basic type of pulverizers.

4.6. Steam power plant:

30 marks, 5 questions, in 1 of 1 papers. Book: Nag Ch. 6, 8.

Most often theory (4 of 5). Subtopics: Improvements; Layout; Boilers; Superheaters and economisers; Boiler efficiency.

Formulas:

Subtopic	Formula	Equation	Used in
Boiler efficiency (8)	Heat from the coal	$Q_{in} = m_f C$	20-21 3(d)
	Boiler efficiency	$\eta_{boiler} = \frac{m_w(h - h_f)}{m_f C}$	20-21 3(d)
	Economiser efficiency	$\eta_{eco} = \frac{m_w c_w(t_2 - t_1)}{m_g c_{pg}(t_g - t_a)}$	20-21 3(d)
	Whole boiler plant	$\eta_{plant} = \frac{m_w(h - c_w t_1)}{m_f C}$	20-21 3(d)
	Gas temperature drop	$\Delta t_g = \frac{m_w c_w(t_2 - t_1)}{m_g c_{pg}}$	20-21 3(d)

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Symbols:

Q_{in} = heat from the coal (kcal/h) ↩

m_f = coal burnt (kg/h) ↩

C = calorific value (kcal/kg) ↩

η_{boiler} = boiler efficiency ↩

m_w = steam (water) per hour (kg/h) ↩

h = enthalpy of the steam (kcal/kg) ↩

h_f = enthalpy of the water entering the boiler (kcal/kg) ↩

η_{eco} = economiser efficiency ↩

c_w = specific heat of water (kcal/kg °C) ↩

t_1, t_2 = water temperature in and out (°C) ↩

m_g, c_{pg} = flue-gas mass (kg/h) and specific heat (kcal/kg °C) ↩

t_g = temperature of the gas entering (°C) ↩

t_a = atmospheric temperature (°C) ↩

η_{plant} = efficiency of boiler and economiser together ↩

Δt_g = fall of the gas temperature (°C) ↩

Questions:

No.	Paper	Marks	Type	Question
Improvements				
1	20-21 1(a)	5	theory	List the factors that contribute to improving the generating capacity of a thermal power station.
Layout				
2	20-21 3(a)	7	theory	Draw the schematics of steam generating part of a coal fire power plant showing all basic components and direction of steam flow.
Boilers				
3	20-21 3(b)	5	theory	Differentiate between water tube boiler and fire tube boiler.
Superheaters and economisers				
4	20-21 3(c)	5	theory	A superheater and an economizer improve the performance of a boiler-How?
Boiler efficiency				
5	20-21 3(d)	8	numerical	The following data is supplied for a boiler plant: Boiler: Mass of coal supplied = 220 kg /hr. Calorific value of coal = 7200 kcal /kg. Mass of feed water = 200 kg/hr *[sic: should be 2000 kg/hr]*. Enthalpy of steam produced = 670 kcal/kg. Economizer: Inlet temperature of feed water = 18°C. Outlet temperature of water = 88°C. Atmospheric temperature = 19°C. Temperature of flue gas entering = 350°C. Mass of flue gases = 4150 kg/hr. Determine the following (i) Efficiency of boiler. (ii) Efficiency of economizer. (iii) Efficiency of whole boiler plant.

4.7. Steam turbines:

6 marks, 1 question, in 1 of 1 papers. Book: Nag Ch. 7.

Most often theory (1 of 1). Subtopics: Impulse and reaction.

Questions:

No.	Paper	Marks	Type	Question
Impulse and reaction				
1	20-21 6(c)	6	theory	Write short notes on: (i) Impulse turbine (ii) Reaction turbines

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4.8. Combined cycle plants:

5 marks, 1 question, in 1 of 1 papers. Book: Nag Ch. 3.

Most often theory (1 of 1). Subtopics: Operation.

Questions:

No.	Paper	Marks	Type	Question
Operation				
1	20-21 1(c)	5	theory	What is meant by combined cycle power generation?

4.9. Hydro plants:

20 marks, 3 questions, in 1 of 1 papers. Book: Nag Ch. 10; Mehta Ch. 2.

Most often theory (3 of 3). Subtopics: Turbines; Civil works; Merits.

Formulas:

Subtopic	Formula	Equation	Used in
Turbines (8)	**Water power**	$P = \rho g Q H$ (W); output $= \eta\, \rho g Q H$	20-21 4(d)

Symbols:

P = power of the water (W) ↩

ρ = density of water (1000 kg/m^3) ↩

g = gravitational acceleration (9.81 m/s^2) ↩

Q = flow (m^3/s) ↩

H = head (m) ↩

η = overall efficiency ↩

Questions:

No.	Paper	Marks	Type	Question
Turbines				
1	20-21 4(d)	8	theory	With diagram, write short notes on Kaplan and Francis turbine.
Civil works				
2	20-21 5(c)	6	theory	Write short note on- (i) Spillways (ii) Surge tanks (iii) Draft tubes
Merits				
3	20-21 6(d)	6	theory	Discuss the advantages and disadvantages of a Hydro-electric power plant.

4.10. Nuclear plants:

5 marks, 1 question, in 1 of 1 papers. Book: Nag Ch. 9; Mehta Ch. 2.

Most often theory (1 of 1). Subtopics: Reactors.

Questions:

No.	Paper	Marks	Type	Question
Reactors				
1	20-21 4(b)	5	theory	Mention the types of reactors used in nuclear power plants.

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5. Theory points:

What a full answer covers, for every theory question, in book order. The link opens the full answer.

No.	Paper	Marks	Subtopic	Points to cover
Load curves and factors				
1	20-21 1(d)	10	Base and peak load	Base load plant: low running cost, steady, reliable, efficient. Peak load plant: quick starting, low capital cost.
2	20-21 2(a)	3	Factors	A high load factor spreads the fixed cost of a plant built for the maximum demand over more kWh; a high diversity factor lets a smaller plant serve the same consumers; both lower the cost per kWh.
3	20-21 2(b)	4	Variable load	Consumers switch loads at will, by time of day, season and weather. Effects: extra (often idle) capacity, higher cost per kWh, varying efficiency, complex operation, stresses on the plant.
4	20-21 4(a)	4	Factors	Plant use factor = energy generated / (plant capacity × hours in operation). Plant utility (utilisation) factor = maximum demand / rated plant capacity.
Economics of generation				
5	20-21 6(a)	7	Depreciation	Straight line: $(P-S)/n$ each year. Diminishing value: a fixed percentage of the remaining value. Sinking fund: a fixed deposit $q=(P-S)\,i/[(1+i)^n-1]$ at compound interest grows to $P-S$ (derived from the geometric series of the deposits).
Tariffs				
6	20-21 5(a)	5	Flat and meter rates	Flat demand rate: bill = Tk per kW of connected load, independent of the energy; no meter; street lighting; invites waste. Straight line meter rate: bill = Tk per kWh × kWh; simple and fair to use, but ignores the maximum demand and the fixed cost.
Fuels and handling				
7	20-21 1(b)	5	Characteristics	Calorific value, moisture, volatile matter, fixed carbon, ash, sulphur, ignition temperature, ease of handling and storage, cost and availability.
8	20-21 6(b)	6	Pulverisation	Pulverised coal burns completely, quickly and controllably, suits any coal and very large boilers, raises efficiency. Ball mill: steel balls in a slowly rotating drum crush the coal; hot air carries the fines through a classifier to the burners.
Steam power plant				
9	20-21 1(a)	5	Improvements	Higher steam pressure and temperature, reheating, lower condenser pressure, regenerative feed heating, economiser and air preheater, good combustion, larger units, a high load factor with good maintenance, combined cycle working.
10	20-21 3(a)	7	Layout	Coal store, crusher, pulveriser and burners; feed water through the economiser to the drum, water walls, drum, superheater, turbine; flue gas past superheater, economiser, air preheater, dust collector, ID fan, chimney; ash removed below.
11	20-21 3(b)	5	Boilers	Water-tube: water inside the tubes, gas outside; high pressure, large output, quick steaming, safer; power stations. Fire-tube: gas inside the tubes in a water shell; low pressure, small output, slow steaming, cheaper.
12	20-21 3(c)	5	Superheaters and economisers	Superheater: dries the steam and raises its temperature, so each kg does more work and the turbine blades are protected from droplets. Economiser: recovers flue-gas heat into the feed water, raising boiler efficiency, saving coal and reducing thermal stress.
Steam turbines				
13	20-21 6(c)	6	Impulse and reaction	Impulse: the whole pressure drop is in the nozzles; the jet pushes symmetrical blades at constant pressure. Reaction: pressure falls in both fixed and moving blades; the moving blades are driven by the reaction of the accelerating steam.

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No.	Paper	Marks	Subtopic	Points to cover
Combined cycle plants				
14	20-21 1(c)	5	Operation	A gas turbine burns the fuel; its hot exhaust raises steam in a heat-recovery boiler that drives a steam turbine; one fuel drives two generators at about 55 to 60 % efficiency.
Hydro plants				
15	20-21 4(d)	8	Turbines	Francis: mixed flow (radial in, axial out) reaction turbine, fixed blades, medium heads. Kaplan: axial-flow propeller reaction turbine with adjustable blades, low heads and large flows, efficient over a wide load range.
16	20-21 5(c)	6	Civil works	Spillway: passes surplus flood water safely past the dam. Surge tank: on the penstock, absorbs water hammer on load drops and supplies water on load rises. Draft tube: widening outlet below a reaction turbine, regains exit energy and lets the turbine sit above the tail race.
17	20-21 6(d)	6	Merits	Advantages: no fuel cost, clean, quick start for peak load, reliable, long life, efficient, multipurpose (irrigation, flood control). Disadvantages: high capital cost, long construction, depends on rainfall, remote sites with long lines, environmental, social and dam-safety risks.
Nuclear plants				
18	20-21 4(b)	5	Reactors	Thermal and fast reactors; in practice PWR, BWR, PHWR (CANDU), gas-cooled (Magnox, AGR, HTGR) and the fast breeder (LMFBR), told apart by fuel, moderator and coolant.

6. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

No.	Paper	Marks	Derivation	Outline
Economic load sharing				
1	20-21 2(c)	10	Incremental rates	Total input $I = I_1 + I_2$ with $P_1 + P_2$ fixed, so $dP_2 = -dP_1$; minimum: $dI = 0$; gives $dI_1/dP_1 = dI_2/dP_2$; second derivative positive; any pair of n units, so all equal λ.

7. Numericals:

Every numerical asked, in book order: the method in one line and the final answer, so a worked attempt can be checked. The link opens the full working.

No.	Paper	Marks	Subtopic	Method	Answer
Load curves and factors					
1	20-21 1(d)	10	Base and peak load	Average load = load factor × peak load; energy = average × 8760 h; diversity factor = sum of maximum demands / peak load; demand factor = peak load / connected capacity.	11.25 MW; 98 550 MWh/year; diversity factor 1.12; demand factor 0.833
2	20-21 4(c)	8	Factors	Energy = maximum demand × load factor × 24 h; plant capacity = average load / capacity factor; reserve = capacity − maximum demand; hours in use from the use factor.	288 MWh/day; reserve 5 MW; 600 MWh/day at full load all day; 360 MWh/day in the 14.4 h scheduled

No.	Paper	Marks	Subtopic	Method	Answer
Economics of generation					
3	20-21 5(b)	5	Cost per kWh	Energy = capacity × load factor × 8760 h; fixed charges = interest and depreciation on the capital + wages and taxes; fixed cost per kWh = fixed charges / energy; add the running cost per kWh.	about 1.98 paisa/kWh (0.98 fixed + 1.0 running)
Tariffs					
4	20-21 5(d)	9	Two-part	Split the costs into running (per kWh) and fixed (per kW); kWh sold after losses; running cost / kWh sold; fixed cost / aggregate demand gives Tk per kW.	Tk 92 per kW per year + 5.29 paisa per kWh
Economic load sharing					
5	20-21 2(d)	8	Two units	Set the incremental costs equal, with $P_A + P_B =$ the load; solve the two linear equations; the loss of equal sharing is $\int$ (incremental cost) dP over the shifted load for each unit.	$P_A = 70$ MW, $P_B = 80$ MW at 15.6 Tk/MWh; equal sharing loses Tk 1.625 per hour
Steam power plant					
6	20-21 3(d)	8	Boiler efficiency	Boiler efficiency $= m_w(h_s - h_{fw})/(m_c C)$; economiser efficiency = heat to water / heat given by the gas; whole plant with the economiser = heat to steam from the inlet water / heat of the coal.	73.48 %; economiser 42.47 %; whole plant 82.32 % (with 2000 kg/h)

8. Diagrams to practise:

Every figure an answer needs, whatever the question's type, in book order. Draw each one from memory, then check it against the solution.

No.	Paper	Kind	Diagram
Load curves and factors			
1	20-21 1(d)	Graph	daily load curve split into base, intermediate and peak load
2	20-21 2(a)	Graph	load factor, average against maximum demand
3	20-21 2(a)	Graph	three consumers peaking at different times
4	20-21 4(c)	Diagram	maximum demand, average load and capacity linked by the factors
Economics of generation			
5	20-21 6(a)	Graph	book value over the life by the three methods
Tariffs			
6	20-21 5(a)	Graph	monthly bill against energy for the two tariffs
7	20-21 5(d)	Diagram	annual cost split into running and fixed charges
Economic load sharing			
8	20-21 2(c)	Graph	incremental cost curves meeting at the economic loading
9	20-21 2(d)	Graph	incremental costs of A and B
Fuels and handling			
10	20-21 6(b)	Diagram	ball mill
Steam power plant			
11	20-21 1(a)	Diagram	steam loop of a thermal station (Rankine cycle)
12	20-21 3(a)	Diagram	steam generating part of a coal-fired plant with steam, water and gas flows
13	20-21 3(b)	Diagram	fire-tube and water-tube boilers
14	20-21 3(c)	Graph	flue gas temperature through the heat exchangers
15	20-21 3(d)	Diagram	where the heat of the coal goes
Steam turbines			
16	20-21 6(c)	Graph	pressure and velocity through impulse and reaction stages

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No.	Paper	Kind	Diagram
Combined cycle plants			
17	20-21 1(c)	Block diagram	combined cycle plant
Hydro plants			
18	20-21 4(d)	Diagram	Francis turbine from above and Kaplan turbine in section
19	20-21 5(c)	Diagram	hydro plant with spillway, surge tank and draft tube
Nuclear plants			
20	20-21 4(b)	Diagram	pressurised water reactor with its two loops

9. Comparisons:

Every comparison asked, in book order, with the key difference in one line. A pair asked in several papers is one row.

Pair	Key difference	Asked in
Load curves and factors		
Base load vs peak load plant	low running cost, runs all the time vs quick starting, low capital cost, runs a few hours	20-21 1(d)
Load factor vs diversity factor	average load / maximum demand, at most 1 vs sum of individual maxima / system maximum, at least 1	20-21 2(a)
Plant use factor vs plant utilisation factor	energy over capacity × operating hours vs maximum demand over rated capacity	20-21 4(a)
Economics of generation		
Straight line vs diminishing value vs sinking fund	equal charges vs large early charges vs equal deposits that earn interest	20-21 6(a)
Tariffs		
Flat demand rate vs straight line meter rate	charged on connected load, no meter vs charged per kWh, ignores maximum demand	20-21 5(a)
Steam power plant		
Water-tube vs fire-tube boiler	water in the tubes, high pressure, large output vs gas in the tubes, low pressure, small output	20-21 3(b)
Superheater vs economiser	heats the steam above saturation vs preheats the feed water with flue gas	20-21 3(c)
Steam turbines		
Impulse vs reaction turbine	pressure drop only in the nozzles vs in both fixed and moving blades	20-21 6(c)
Combined cycle plants		
Combined cycle vs simple steam plant	gas turbine exhaust reused, 55 to 60 % efficient vs one cycle, about 35 to 40 %	20-21 1(c)
Hydro plants		
Francis vs Kaplan turbine	mixed flow, fixed blades, medium head vs axial flow, adjustable blades, low head, large flow	20-21 4(d)
Hydro vs thermal plant	no fuel cost, high capital cost, depends on rain vs fuel cost, lower capital cost, near the load	20-21 6(d)
Nuclear plants		
PWR vs BWR	water kept liquid at high pressure, separate steam generator vs water boils in the core, steam straight to the turbine	20-21 4(b)

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10. Repeated questions:

No repeats among these papers.

11. Question style:

How each topic was asked, as a number of questions.

Topic	Theory	Derivation	Numerical
Load curves and factors	4	·	1
Economics of generation	1	·	1
Tariffs	1	·	1
Economic load sharing	·	1	1
Fuels and handling	2	·	·
Steam power plant	4	·	1
Steam turbines	1	·	·
Combined cycle plants	1	·	·
Hydro plants	3	·	·
Nuclear plants	1	·	·

12. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

No.	Topic	Marks	Covered so far
1	Steam power plant	30	20 %
2	Load curves and factors	29	39 %
3	Hydro plants	20	53 %
4	Economic load sharing	18	65 %
5	Tariffs	14	74 %
6	Economics of generation	12	82 %
7	Fuels and handling	11	89 %
8	Steam turbines	6	93 %
9	Combined cycle plants	5	97 %
10	Nuclear plants	5	100 %